



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD MPP.4

Math is Ingenuity

Lesson 10-4 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can use ratios and models to explain how the Penny Farthing and gear-driven bicycles were ingenious solutions to a real problem.

Language Objective: I can explain a gear ratio using the words rotation, teeth, and 'for each ____, the wheel turns ____.'



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

ingenuity

rotation

gear

ratio

equivalent ratios



Tap any word to see its meaning and a picture.

1

Coming up with a different, sometimes unique solution is

2

One full turn of a wheel, gear, or pedal is a

3

A toothed wheel that passes motion through a chain is a

4

A comparison of two quantities, like pedal turns to wheel turns, is a

5

Ratios that make the same comparison with different numbers are

Write About the Math

How will your future vehicle be powered, and what math will be used in its design or use?

¿Con qué funcionará tu vehículo del futuro, y qué matemáticas se usarán en su diseño o su uso?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **ingenuity**
ingenio

☐ **rotation**
rotación

☐ **gear**
engranaje

☐ **ratio**
razón

☐ **equivalent ratios**
razones equivalentes

Model: One pedal rotation makes one wheel rotation, so a bigger wheel's longer perimeter carries the bicycle farther with each pedal push. — A strong answer connects the 1-to-1 ratio to the wheel's perimeter, explaining a bigger wheel travels farther per pedal push. A weak answer just says 'bigger is faster' without naming the ratio or perimeter.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Describe a future mode of individual transportation. A complete answer says how it is ____, names a ____ (or rate) that shapes it, and explains where that math appears in the vehicle's ____.**
 - **My answer is ____ because ____.**
Mi respuesta es ____ porque ____.
-
-

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.

- **My evidence is** ____, **so** ____.

Mi evidencia es ____, *entonces* ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** ingenuity
2. **Notes 2:** rotation
3. **Notes 3:** gear
4. **Notes 4:** ratio
5. **Notes 5:** equivalent ratios
6. **Watch for this mistake:** *Thinking a rear gear with MORE teeth makes the wheel spin more — it is the opposite: the 32-tooth gear gives only 1.5 rotations per pedal rotation, while the 12-tooth gear gives 4.*
7. **Try It 1:**
8. **Try It 2:**